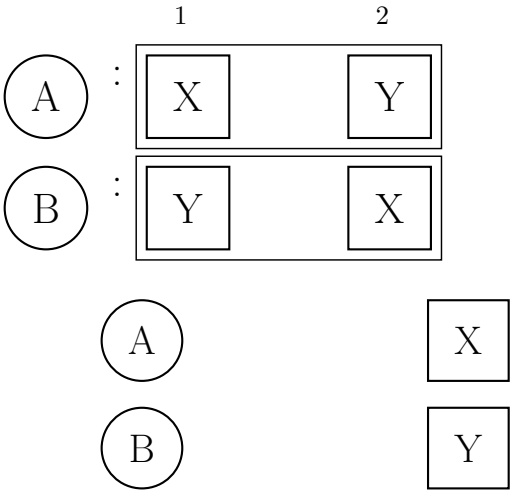
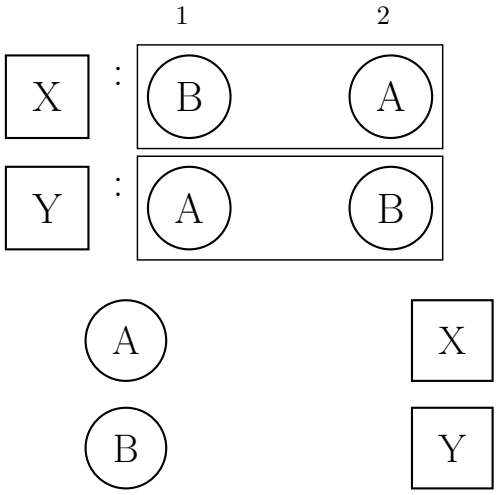
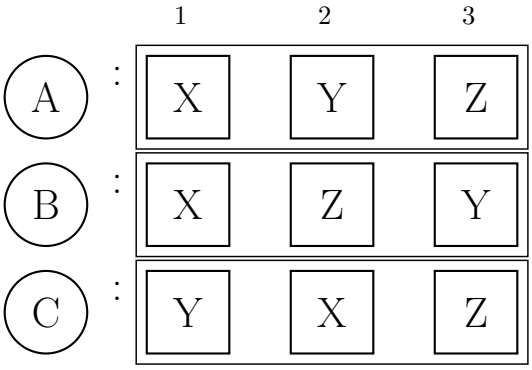
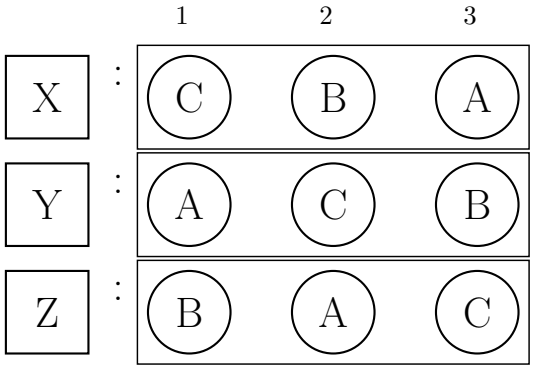
Pair each circle with one square, using every letter once. Each list runs from first choice to last, with no ties. A pairing is stable if no two letters in different pairs both prefer one another to their current partners.

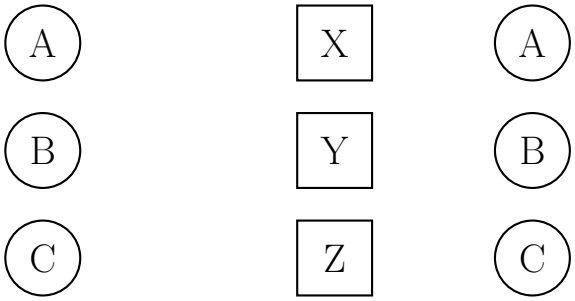
Problem 1: Find all stable pairings for each set of lists.

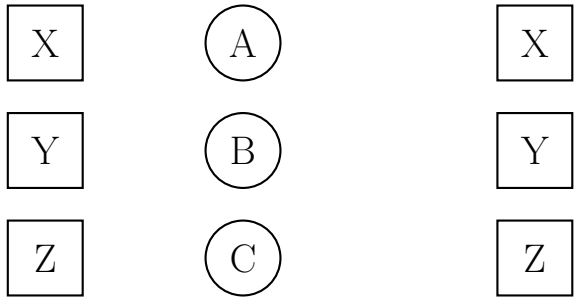








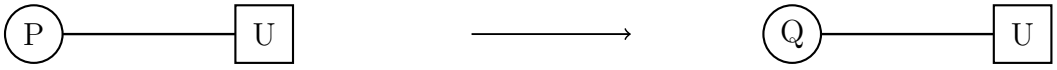




Problem 2: Use these rules with A, B, C, D asking. Show or record the requests and final pairs. Run them again with W, X, Y, Z asking. Does changing the asking side change the final pairing?

Begin with everyone free. A free asker asks its highest choice that it has not asked before. A free receiver holds its asker. A receiver already holding someone keeps whichever of the two it prefers and releases the other. Holds are temporary. Stop when every asker is held.

One step with other letters: U ranks Q ahead of P. Free Q asks U.



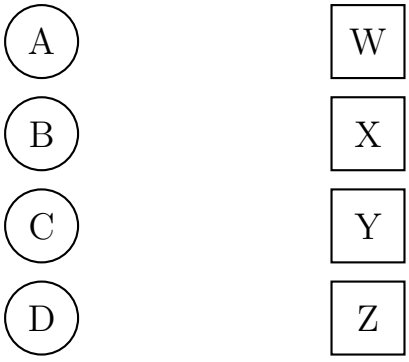
Before: P held; Q free. After: Q held; P released.

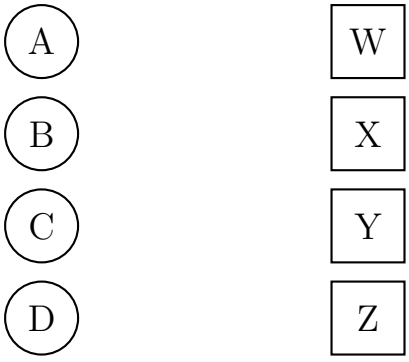
	1	2	3	4
A	X	W	Z	Y
B	X	Y	W	Z
C	X	Y	Z	W
D	X	Y	W	Z

A, B, C, D asking

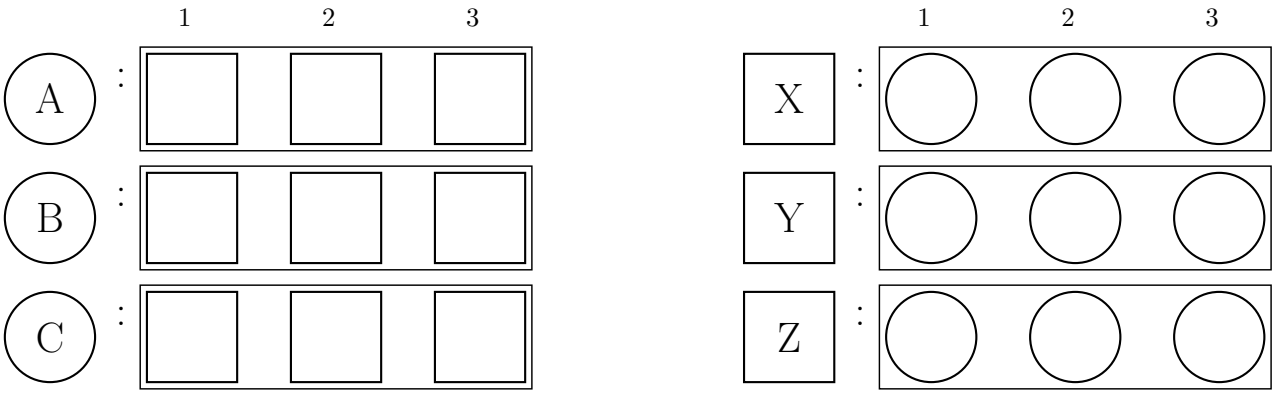
	1	2	3	4
W	B	C	D	A
X	D	B	A	C
Y	B	C	A	D
Z	B	D	A	C

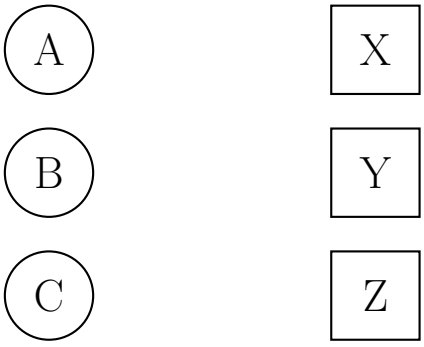
W, X, Y, Z asking





Problem 3: Change the rule: a receiver must keep the first asker and reject every later asker. Choose complete lists for A, B, C and X, Y, Z that make these changed rules end with an unstable pairing. Let A, B, C ask; record the requests and two letters in different pairs that both prefer each other.





Problem 4: Four letters on each side follow the asking rules from Problem 2. Give a number of requests that can never be exceeded, and explain why the rules must finish with everyone paired. Would your argument work with ten letters on each side?

Problem 5: When the asking rules from Problem 2 finish, could two letters in different pairs both prefer one another to their final partners? Explain why your answer holds for every set of complete lists with no ties and equal-sized sides.

Problem 6: Choose complete lists for A, B, C and X, Y, Z and a stable pairing in which as many of the six letters as possible get their last choice. Explain why a larger number is impossible.

